Example 6.39

Let $u = \log \frac{x^4 + y^4}{x + y}$. By using Euler's theorem show that $x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = 3$.

Solution:

$$u = \log \frac{x^4 + y^4}{x + y}$$

$$e^u = \frac{x^4 + y^4}{x + y} = f(x, y) \quad \ldots (1)$$

Consider $f(x,y) = \frac{x^4 + y^4}{x + y}$

$$f(tx, ty) = \frac{t^4x^4 + t^4y^4}{tx + ty} = t^3\left(\frac{x^4 + y^4}{x + y}\right) = t^3 f(x, y)$$

$\therefore$ f is a homogeneous function of degree 3.

Using Euler's theorem we get,

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = 3f$$

Consider $f(x,y) = e^u$

$$x \cdot \frac{\partial e^u}{\partial x} + y \cdot \frac{\partial e^u}{\partial y} = 3e^u$$

$$\therefore \quad e^u x \cdot \frac{\partial u}{\partial x} + e^u y \cdot \frac{\partial u}{\partial y} = 3e^u$$

$$x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = 3$$

Exercise 6.4

1. If $z = (ax + b)(cy + d)$, then find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
2. If $u = e^{xy}$, then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = u\left(x^2 + y^2\right)$.
3. Let $u = x\cos y + y\cos x$.
   Verify $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$.
4. Verify Euler's theorem for the function $u = x^3 + y^3 + 3xy^2$.
5. Let $u = x^2y^3\cos\left(\frac{x}{y}\right)$. By using Euler's theorem show that $x.\frac{\partial u}{\partial x} + y.\frac{\partial u}{\partial y} = 5u$.

6.5 Applications of Partial Derivatives

In this section we solve problems on partial derivatives which have direct impact on Industrial areas.

6.5.1 Production function and marginal productivities of two variables

(i) Production function:

Production P of a firm depends upon several economic factors like capital (K), labour (L), raw materials (R), machinery (M) etc... Thus $P = f(K, L, R, M, \ldots)$ is known as production function. If P depends only on labour (L) and capital (K), then we write $P = f(L, K)$.

(ii) Marginal productivities:

Let $P = f(L, K)$ be a production function. Then $\frac{\partial P}{\partial L}$ is called the **Marginal productivity of labour** and $\frac{\partial P}{\partial K}$ is called the **Marginal productivity of capital.**

Euler's theorem for homogeneous production function $P(L,K)$ of degree 1 states that $L\frac{\partial P}{\partial L} + K\frac{\partial P}{\partial K} = P$

6.5.2 Partial elasticity of demand

Let $q = f(p_1, p_2)$ be the demand for commodity A, which depends upon the prices.

p_1 and p_2 of commodities A and B respectively.

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